MediCompare

A neutral, source-traceable medicine price information system for India, with human-verified prescription reading

Project report

November 2026

Information only, not medical advice. This is a prototype. It is not deployed, has not been tested with users, and no regulatory compliance is claimed. Real prices are NPPA legal maximums. Shop prices in this build are fictional demo data.

Abstract

Medicine prices in India vary widely across brands, yet choosing a cheaper product wrongly can cause harm. This project designs and builds MediCompare, a prototype that treats this risk as the central requirement. Optical character recognition (OCR) only proposes candidates: a person must confirm each medicine, and the server refuses price comparison until they do. Products are compared only when a deterministic rule proves them identical on ingredient, salt form, strength, dosage form, route and release type. 352 official NPPA price rows were imported with row-level provenance. Two OCR engines were evaluated on 600 synthetic prescriptions. At thresholds calibrated on validation data, the selected engine made 1 false match(es) in 67 suggestions on the test split (95% upper bound 8.0%), at 17.4% coverage. The system passes 17 of 17 automated gates, including 99 safety tests. The main limitations are that results come from synthetic handwriting, that no real shop prices were available, and that no user testing has taken place.

1. Introduction

The same molecule can be sold at very different prices under different brands. Patients and caregivers also struggle to read handwritten prescriptions, distinguish brand from generic names, assess whether a price is trustworthy, and find Jan Aushadhi options. Tools that automate these tasks carry a specific danger: a confident, wrong identification may lead someone to the wrong medicine.

Objectives: to build a working application that searches medicines, shows only sourced prices, surfaces lower-cost options with their differences, supports Jan Aushadhi discovery, and reads prescriptions with mandatory human verification, and to measure how it behaves rather than assert it.

Scope: MediCompare is not a pharmacy, prescriber, diagnostic tool, dosing or interaction advisor, or emergency service.

2. Related work

Observations were made from public pages and search results in November 2026. They are brief, and are not a full feature audit. Online pharmacies such as Tata 1mg, PharmEasy and Apollo 24|7 sell medicines and accept prescription uploads to fulfil orders, so they are not neutral. Two government tools are closest in spirit: NPPA's Pharma Sahi Daam offers official price lookup, and Janaushadhi Sugam offers a Kendra locator and generic versus branded comparison. MediCompare does not have better data than these. Its contribution is a design pattern: enforced human verification, and refusal to compare what cannot be proven comparable.

3. Requirements and safety rules

- Never guess an unreadable medicine, and never infer dose, frequency, duration, strength or route.
- Never let an unverified identity reach comparison, enforced on the server.
- Never treat similar names as equivalent.
- Present alternatives only as options to discuss with a doctor or pharmacist.
- Interrupt on emergency language with India's 112 number.

- Never present a ceiling price as a retail price, and never imply stock.
- Label all synthetic data, and never fabricate results.

4. System design

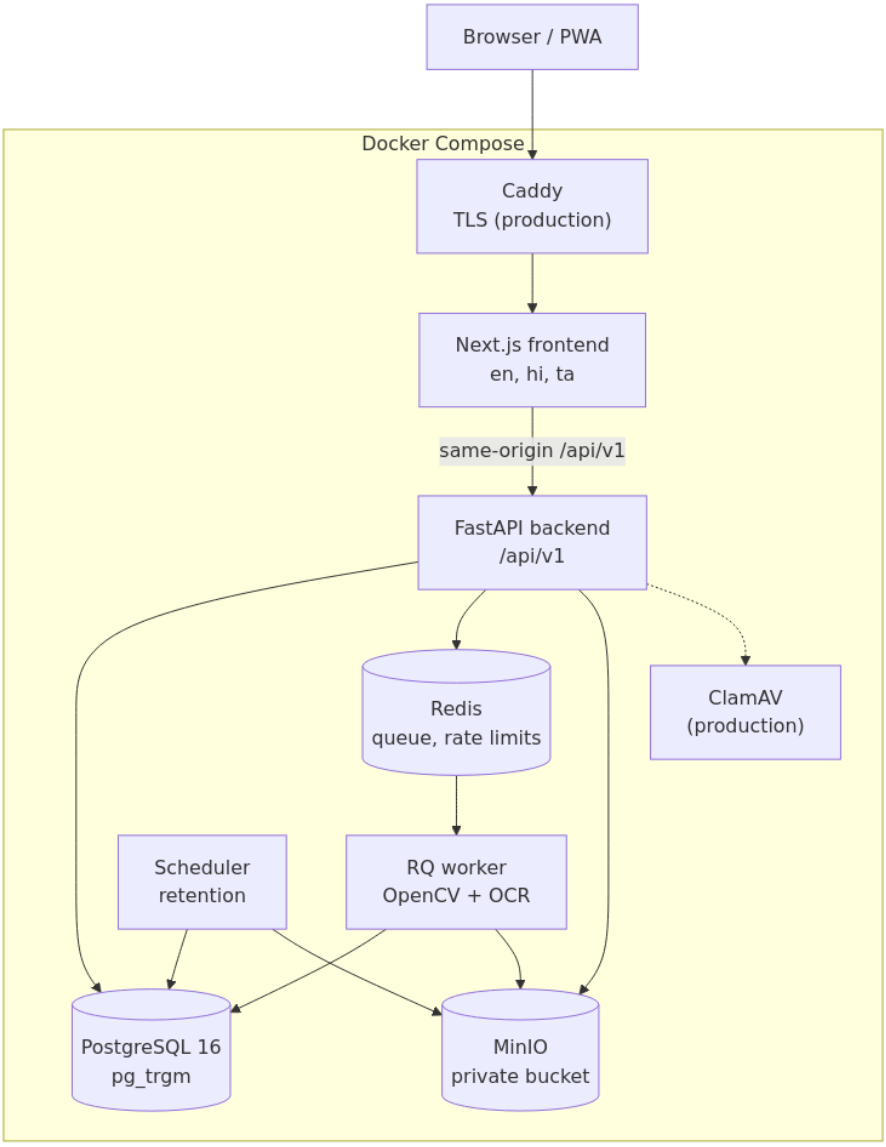


Figure 1. Container view.

4.1 Equivalence

One pure function compares two products attribute by attribute and returns every difference. Names are not an input, so look-alike names cannot produce a match. Strengths are converted to canonical units, so 250 mg/5 mL equals 50 mg/mL. An unstated salt form cannot prove a match. This has a real cost: NPPA's ceiling lists do not state salts, so those rows rarely compare per unit with other products.

4.2 Prescription pipeline

Uploads are validated by magic bytes and limits, and re-encoded from pixels to remove metadata. Images are deskewed and enhanced with OpenCV. An OCR engine produces text with boxes and quality. Deterministic extraction drops instruction-like text and personal details, then records each field as visible, unclear or unknown. Matching returns explained candidates, and never selects one.

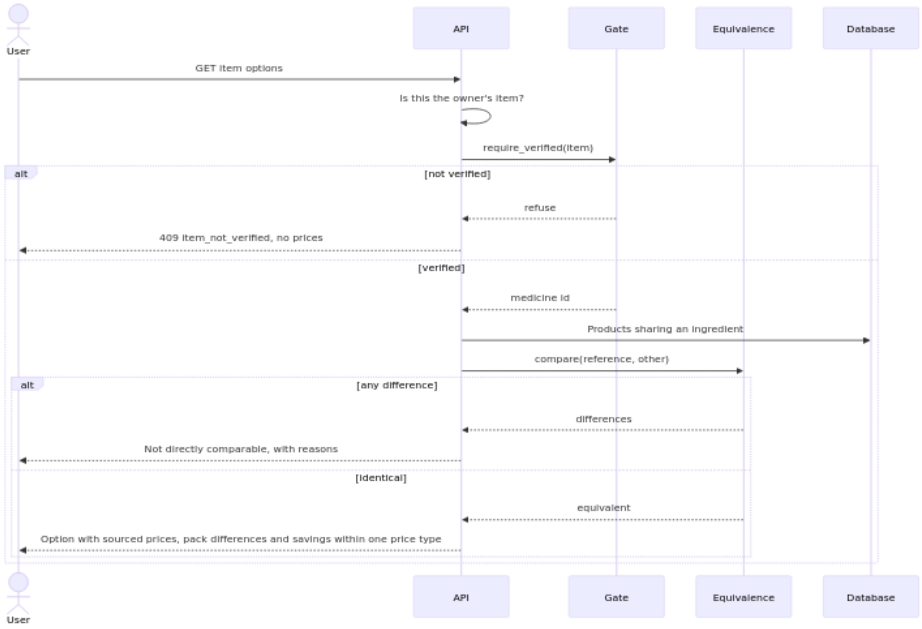


Figure 2. Comparison through the verification gate.

4.3 The gate

Price and option routes for a prescription item require status verified, a confirmed medicine and a verifier type. A database constraint rejects a verified row without these. The gate takes no confidence score, so a score can change a label but never access.

4.4 Data

NPPA's copyright policy permits reproduction with accuracy and acknowledgement, so its therapeutic-category lists and text-based Authority minutes were parsed from PDF. Scanned minutes were deliberately not read by OCR, since a misread digit would appear as an official price. Imports pass through staging and validation: 352 rows were published and 38 wait in a review queue with stated reasons. The Jan Aushadhi portal requires permission to reproduce its data, so that data was not bundled, and demo data stands in. Demo brands all start with "Demo", and a data-quality check enforces this.

4.5 Privacy and security

Controls include Argon2id, httpOnly cookies, CSRF tokens, rate limits, lockout, a nonce-based Content Security Policy, private object storage, 7-day retention, data export and hard deletion. A STRIDE threat model is in docs/THREAT_MODEL.md. Known gaps include a single database role, encryption at rest left to the host, and no penetration test.

5. Evaluation

5.1 Method

A seeded generator rendered 600 prescriptions (360 train, 120 validation, 120 test) using eight open-licence handwriting fonts, with rotation, perspective, shadow, blur, noise and low resolution. Hard cases included look-alike brands, missing strengths, fixed-dose combinations, modified release, illegible scribbles and injected instructions. Scoring was strict: any suggestion on a line with a missing strength, an illegible span or non-medicine text counts as a false match. Thresholds and engine choice used validation data only.

5.2 Results

Test split	tesseract (selected)	rapidocr
Character error rate	29.4%	20.7%
Word error rate	48.4%	34.7%
Medicine name F1	30.4%	42.1%
Top candidate correct	36.1%	54.5%
False-match rate, no thresholds	21.7%	20.1%
False-match rate, calibrated	1.5% (1/67)	2.2% (2/91)
95% upper bound	8.0%	7.7%
Coverage of readable lines	17.4%	23.4%
Abstention rate	83.1%	77.1%
Latency per page, p50	1.04 s	2.63 s

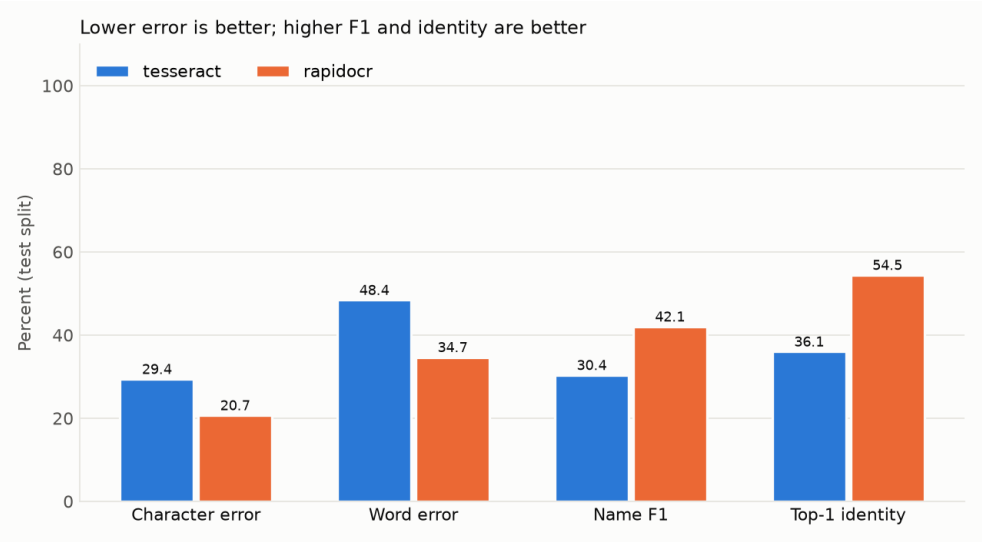


Figure 3. Engine comparison on the test split.

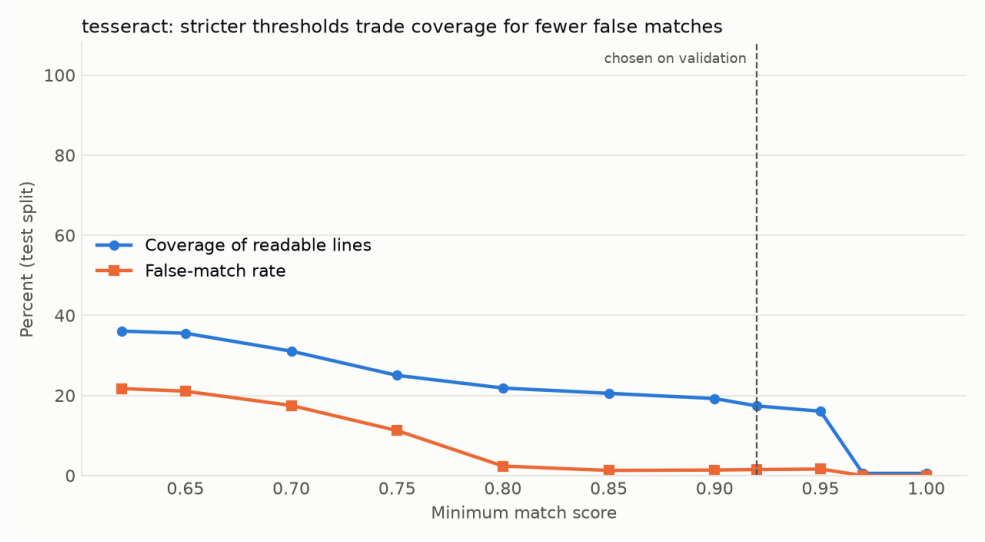


Figure 4. Threshold trade-off for the selected engine.

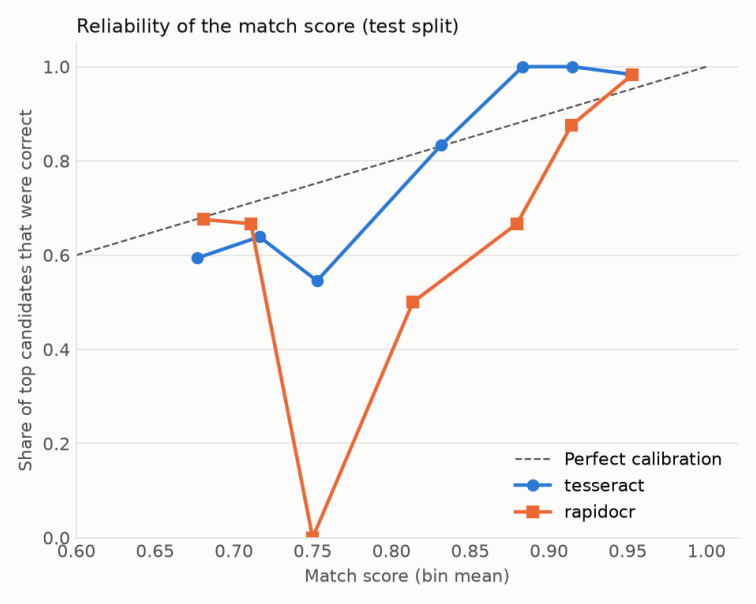


Figure 5. Reliability of the match score.

5.3 Discussion

Without thresholds, about one top candidate in five was wrong, even on this easy data. This supports the decision that software must not select a medicine. Calibration reduced false matches sharply, at the cost of coverage: most lines receive no suggestion. tesseract was selected by the rule fixed before scoring, but its false-match rate cannot be statistically distinguished from rapidocr's, which reads text better. The match score is not a probability, and should not be presented as one.

Threats to validity. Fonts are far more regular than handwriting, the templates are clean, and only catalogue names appear. With under 100 suggestions per engine, rare serious errors cannot be excluded. Nothing here measures whether people confirm correctly when shown a suggestion. **These results do not show real-world accuracy.**

6. Testing and verification

Gate	Result
Backend tests	295 passed; 98.3% line coverage; 97.3% on core modules
Safety invariant suite	99 passed, including direct API bypass attempts
Frontend tests	65 passed; 97.91% line coverage
End-to-end, with axe	40 runs across desktop and mobile viewports; no serious or critical violations
Load test (local, one laptop)	Search p50 150 ms, p95 530 ms, p99 1100 ms; 6885 requests, 0 failures, 57.7 requests/s
All gates	17 of 17 passed. See VERIFICATION_REPORT.md

Testing found real defects, including partial words classed as exact search matches, demo accounts unable to sign in, rate limits that would have grouped all users as one address, and virtual environments being copied into container images. One accepted security finding remains: an old PCRE2 library bundled in two Python packages, documented with its rationale. The load test ran with raised rate limits, client and server sharing one machine, so it does not indicate production capacity.

7. Limitations

- No real shop prices exist in this build. Real prices are NPPA legal maximums, which vary little between companies.
- Real data covers five ceiling-price categories, dated March 2026 and without salt forms, and new drugs from four sets of minutes.
- OCR results are from synthetic data only. No real prescriptions were used.
- The look-alike check is a string-similarity screen, not a clinical list.
- Hindi and Tamil text is AI-written and awaits native-speaker review.
- No user research, pharmacist testing, screen-reader user testing, or low-end device testing has been done.
- No legal review has been done. Whether this is medical device software is an open question.
- The system is not deployed, and its CI workflow has not run on GitHub.

8. Conclusion and future work

MediCompare shows that a medicine price tool can be designed so that safety rules are enforced by the system, and that those rules can be tested. The evaluation supports mandatory human verification, even on favourable data. It does not show that the product helps anyone yet, because that needs real data and real users.

- Obtain real shop prices and PMBI permission.
- Obtain legal review, beginning with medical device classification.
- Run a consented, ethics-approved evaluation on real prescriptions (protocol in docs/EVALUATION.md).
- Test with pharmacists and patients, measuring whether people catch wrong suggestions.

- Review translations with native speakers, and replace the similarity screen with a clinical list.

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